Assignment

CSE-812 RF Communication System Design

Instructions:

- General Instructions
- Show all calculation steps clearly for numerical parts.
- Use proper SI units and engineering notation.
- Clearly state any assumptions made.
- For theoretical questions, answers should demonstrate technical understanding and engineering interpretation, not only definitions.
- Numerical answers without supporting calculations may receive reduced marks.

Question 1 – RF Impedance and Frequency-Dependent Behavior [10 Marks]

(a) Theoretical

Explain the difference between resistance, reactance, and impedance in an RF circuit. Discuss how the impedance of an ideal resistor, capacitor, and inductor varies with frequency.

Also explain why practical RF components may deviate from their ideal behavior at high frequencies due to parasitic effects.

(b) Numerical

A series RF circuit consists of R = 50 Ω, L = 15 nH, and C = 2 pF, and operates at f = 1.5 GHz.

Calculate:

1. Inductive reactance, X_L
2. Capacitive reactance, X_C
3. Total complex impedance, Z
4. Magnitude of impedance, |Z|
5. Phase angle of the impedance
6. State whether the circuit is predominantly inductive or capacitive.

Question 2 – Electromagnetic Wave Generation and RF Propagation [10 Marks]

Using the concepts of Maxwell's equations, explain how a time-varying electrical signal applied to an antenna results in the generation and propagation of an electromagnetic wave.

Your answer should include discussion of:

- Relationship between time-varying electric and magnetic fields
- Role of an antenna in RF transmission and reception
- Transverse nature of electromagnetic waves
- Reflection
- Diffraction and refraction

- Multipath propagation
- Constructive and destructive interference
- RF fading

Relate these effects to the behavior of RF signals in a practical wireless environment.

Question 3 – RF Power, dB and Free-Space Propagation [10 Marks]

(a) RF Power and Attenuation

A sinusoidal RF signal having an RMS voltage of V_{in} = 2 V is applied to a 50 Ω RF system. The signal subsequently passes through a 13 dB attenuator.

Calculate:

1. Input RF power in watts
2. Input RF power in dBm
3. Output RF power in dBm
4. Output RF power in watts
5. Output RMS voltage
6. Percentage of input power remaining at the output

Use the appropriate relationships between power, voltage, resistance, dB, and dBm.

(b) Free-Space Propagation

An isotropic RF transmitter radiates P_{TX} = 20 W. Determine the power density at d_1 = 100 m and d_2 = 1 km.

Also calculate the reduction in power density, expressed in dB, when the distance increases from 100 m to 1 km.

Briefly explain the physical reason for the calculated reduction in signal level.